

A Data-Driven Framework for Crop Price Prediction Using ML, Statistical, and Hybrid Ensemble Models

Manimegalai R.^{1*}, Logendar G.¹, Srirengapriya G.¹ and Ayesha S.K.²

¹*Department of Computer Science and Engineering, PSG Institute of Technology and Applied Research, Coimbatore, India*

²*Department of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India*

Abstract

Crop price prediction is one of the big areas of interest in agricultural economics, concerning farm planning and decision-making for the farmer, stakeholder, or policymaker. In this work, various machine learning models are developed to predict crop prices accurately using the historical dataset. The proposed methodology in this paper identifies the robust crop price prediction model. Models are built using random forest regression, long-short-term memory (LSTM), and statistical time series models such as autoregressive integrated moving average (ARIMA) and seasonal autoregressive integrated moving average (SARIMA). The models built using the above mentioned algorithms are compared on the basis of their accuracy and robustness. The proposed methodology involves gathering and pre-processing the dataset consisting of crop prices from historical data. Tasks such as feature engineering, model training, hyper-parameter tuning, ensembling, and evaluation are carried out in order to identify the robust model for crop prediction. Experimental results demonstrate the validity of the proposed methodology in terms of predicting crop prices accurately, with low error rate. The proposed approach contributes to the advancement in the methods of agricultural forecasting and improves insights into informed decisions in crop marketing and management. This will thereby offer several avenues for follow-up research, wherein, through considerations of weather condition, market dynamics, and crop-specific factors, it may be possible to make improved predictions. Random forest model achieved the highest accuracy, outperforming traditional

*Corresponding author: drmm@psgitech.ac.in

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statistical models. LSTM model effectively captured temporal dependencies, whereas XGBoost showed overfitting despite high training accuracy. The hybrid stacked ensemble model further improved prediction accuracy by combining multiple algorithms, enhancing crop price forecasting.

Keywords: Machine learning, ensembling techniques, feature engineering, hyper-parameter tuning

17.1 Introduction

From fundamental statistical techniques to intricate econometrics integrated with artificial intelligence (AI), the methods of crop price forecasting have certainly advanced. Earlier forecasts depended on analyzing the past and available data whereas machine learning (ML) came into the picture in the early 2000s with support vector machine (SVM) and random forest models performing better than the traditional methods [17]. This said shift reflects remarkable improvement in predictive accuracy.

The challenges in India's agricultural sector are many: inaccurate forecasting, volatile pricing, and asymmetric information flow. Employing half of the nation's workforce and contributing significantly to GDP, the sector is economically vital; however, it currently faces dire challenges. Indian farmers, more specifically, the 86% of them who are smallholders, suffer massive losses due to market uncertainties and price meltdowns. This forces uninformed decisions about crop selection and market strategies that exacerbate the problem. Limited access to sustainable markets, surpassing intermediaries, and supply chain fragmentation all prevent farmers from achieving mortgaged sustainable profitability. Cropping and cultivation decisions become economically motivated while the economically motivated strategies fail commercially. Modern monitoring systems including Kisan Suvidha and the Uzhavan app boast features targeted toward addressing farmer woes at different levels, although they all suffer from under-performing data integration and ML implementation. Unrestricted scalability and underdevelopment leads to poor model construction, real-time data processing, and region agnostic deployment claiming to cater these Mauritius-based loopholes local market modular architecture, and versioning, insufficient predictive frameworks, ensemble learning, and advanced applicable methodologies leave the platforms lacking adaptable local market dynamics.

This study seeks to solve the problem by implementing ARIMA, SARIMA, long-short-term memory (LSTM), random forest, XGBoost,

and advanced ensemble learning such as stacked and hybrid stacked ensemble models to accurately predict crop prices. Random forest was the best at individual accuracy with 98.32% testing accuracy. However, the greatest achievement was the hybrid stacked ensemble model with a linear regression meta-model, attaining a remarkable 97.24% testing accuracy. The research offers a comprehensive framework to integrate advanced ML technologies with agriculture 4.0 platforms, which changes the role of these tools from mere informational resources into integrated, intelligent decision support systems that assist farmers in choosing crops, planning production schedules, and pricing—notably elevating agricultural efficiency and farmer welfare throughout India.

17.2 Literature Survey

Sopee Kaewchada *et al.* have suggested the random forest model for forecasting vegetable prices of Nakhon Si Thammarat Province in Thailand by comparing the outcomes of different crops [1]. It utilized monthly climate data and year 2011–2020 average monthly vegetable prices. The performance of the model is verified using mean absolute percentage error, root mean squared error (RMSE), and mean absolute error (MAE). Results in [1] show that incorporation of soil conditions, insects, and vegetable types into a model would provide significant value-addition to prediction accuracy. Pandit Samuel *et al.* in [2] suggest a ML system for forecasting crop price using data analytics methods and a combination of ML algorithms including XGBoost, logistic regression, decision trees, and neural networks. XGBoost model offers the highest crop-price-prediction with maximum accuracy over the other ML models verified on the performance basis by the RMSE. The possible inclusion of additional factors, including market trends and up-to-date data, raised the level of accuracy in predictions. Nitesh Singh *et al.* have tackled one such problem of predicting crop prices accurately and resolved it through ML methods like regression models, time series prediction, ensemble methods, and deep learning for the prediction of crop prices [3]. Random forest was more accurate and robust than other models, decision tree regressor had an overfitting characteristic with the data, and a linear regressor gave a baseline but struggled with intricate interdependencies within the data.

Neeraj Soni *et al.* have designed a ML model that can predict crop prices in India using the decision tree regressor and random forest models by analyzing historical rainfall. Crop price prediction up to 12 months ahead was done for effective management of fluctuation [4]. The study reported

accuracy levels as it is based on the difference between predicted and actual prices, showing that the decision tree regressor is more accurate than random forest. It emphasized how having data of improved quality, with real-time updates, and other market factors added will make prediction even more robust and reliable. Murugesan *et al.* have proposed three variations of hybrid ensemble model which put together ARIMA with support vector regression (SVR) to obtain more accurate crop price forecasting: i) using SVR on the residuals of ARIMA, ii) using ARIMA predictions as an input to SVR, and iii) using its predictions and residuals as inputs for SVR [5]. The third model assessed on the data of coconut price represented the highest accuracy when compared to other models. Results of the study also implied that, to achieve better performance concerning the prediction, it might consider including more variables and multi-commodity applicability.

Konstantinos G. Liakos *et al.* have investigated the application of ML in crop, livestock, water, and soil management [6]. In their work, the efficiency of artificial neural network (ANNs), SVMs, and decision trees in yield estimation, disease identification, and resource management has been noted. Crop and soil management have been aided by ANNs and random forest for precise estimations, whereas water and livestock management have been performed using SVMs and regression models. The article has also highlighted the significance of data through sensors and big data analytics in enhancing agricultural productivity but has fallen short of including real-time case studies and advances in deep learning. The authors have proposed combining IoT, real-time data, and deep learning models for increased precision in agriculture. R. Rohith *et al.* have suggested a mechanism based on decision tree regression and SVR to forecast crop prices based on variables such as rainfall, minimum support price, and wholesale price index [7]. The need for price forecasting to enable farmers to plan effectively and minimize financial risks has been emphasized. Although the model has offered a 12-month prediction and a web-based interface for access, external economic factors and real-time market trend incorporation have been absent. Decision trees have been shown to correctly forecast price trends, but precision would have been enhanced with more variables such as fertilizer recommendations, which have been proposed as a future addition.

I. Ghutake *et al.* have explored numerous ML methods for predicting crop prices, determining the most appropriate algorithm to enhance forecast accuracy [8]. The research has centered on examining past price data and outside factors such as climate and market demand. Although their methodology has enhanced predictive accuracy, the model has been

limited by the absence of real-time market movement and extraneous economic factors, which may have facilitated greater precision in the system. Zhang *et al.* have created a selection framework for best forecasting models according to time series characteristics [9]. Their research has tested various forecasting horizons to identify the best model for agricultural price forecasting. The methodology has proved to show considerable improvement in forecast precision, especially for short- and medium-term forecasts. Their model has, however, failed to capture unexpected market shocks and price instability due to unforeseen external events like policy twists, global trade hindrances, and extreme climatic changes.

M. Subhasree and C. A. Priya have utilized time series forecasting to forecast vegetable prices using past trends [10]. Their findings have given insight into price variation and seasonal changes, enabling effective planning in markets as well as inventory management. But the research has been constrained by its use of historical data in isolation without blending it with external economic data like inflation levels, transport prices, or consumer demand trends, which might have sharpened the precision of the price forecasts. S. Bayona-Oré *et al.* have investigated the use of ML algorithms to enhance price forecasting for agri-commodities [11]. The study has proven that ML models have the capability to identify intricate patterns in price movements and perform better than conventional statistical methods. Still, although promising, the study has failed to integrate real-time data sources such as global commodity markets, government policies, and logistics of supply chains, which could have enhanced the system to be more dynamic and robust for forecasting.

B. Sahithi *et al.* have suggested a supervised ML-based model for forecasting crop prices by utilizing decision tree regression [12]. The system has used variables such as rainfall, wholesale price index, and minimum support price to create a 12-month forecast. The research has successfully overcome the issue of price volatility in agriculture but has been hindered because there were no other influencing factors like global trade regulations, outbreaks of pests, and inflation in the economy, which could have further improved the forecasting model. M. Kaur *et al.* have examined the data mining methods of predicting crop prices with emphasis on the contribution of ML models to agricultural market intelligence [13]. Their research has examined some of the computational methods, such as decision trees, SVMs, and ensemble-based learning techniques, to identify their effectiveness in price forecasting. Although the study has been able to illustrate the benefits of ML methods, it has been constrained by access to and the nature of historical datasets for agricultural prices, which might have created biases in the forecasts.

M. Mukhtar *et al.* have constructed a hybrid ML model based on strong regression methods to improve agricultural sustainability and food security [14]. Their research has stressed the use of predictive analytics for maximizing crop prices and production planning. Although their hybrid method has obtained better accuracy, the research has not considered unforeseen geopolitical forces and natural disasters, which have dramatic effects on crop prices in the world market. K. M. Sabu and T. K. M. Kumar have made use of predictive analytics to predict the areca nut prices, a prominent agricultural product in Kerala [15]. These two researchers have made use of historical prices and regional economic indicators in order to forecast the market trends. The research has been informative to farmers and traders but has been limited by the lack of macroeconomic determinants like currency exchange rates and global trade policies, which affect farm prices at a wider level.

A. Darekar and A. A. Reddy have examined paddy price movements by applying statistical forecasting techniques [16]. Their research has shown that price forecasting is essential to stabilize the agricultural economy and minimize the financial risks of the farmers. Although their methodology has been successful for short-term prediction, it has not completely explained the effects of climate variability, pest attacks, and government subsidy policies that are key determinants of long-term paddy price. A weather forecasting model for wind speed data and prediction using neural network based algorithms are discussed in [20, 22]. A data prediction scheme using time-series classification is presented in [21].

17.3 Methodologies

The architecture of the crop price prediction system under consideration is presented in Figure 17.1. The agricultural yield statistics and environmental data were gathered and were fed into a data preprocessing pipeline where missing values were handled, outliers were identified and discarded and feature extraction methods were used to improve the quality and predictive capabilities of the data. The preprocessed data was thereafter split into training and test subsets to enable model development and evaluation. Multiple ML models, including ARIMA, SARIMA, LSTM, random forest, and XGBoost, were trained on the data. The base models were then combined in a stacking ensemble framework, wherein their output was merged and smoothed by a meta-model to enhance predictive performance. The performance of the last predictions was analyzed through common metrics, such as mean squared error, RMSE, coefficient of determination (R^2),

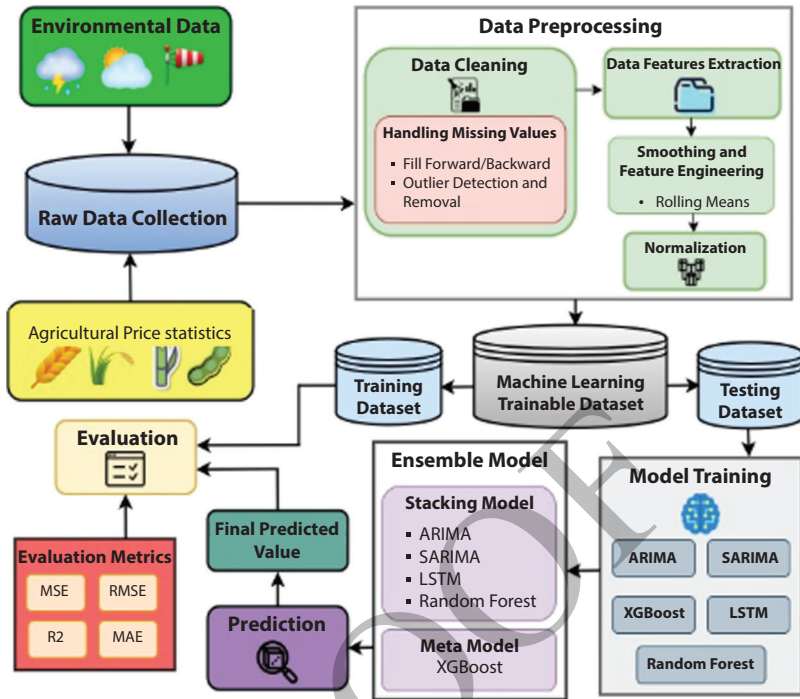


Figure 17.1 Block diagram of the proposed crop price prediction system.

and MAE. The addition of feature engineering, hyper-parameter optimization, and ensemble strategies did a lot to improve the strengths and dependability of the suggested system.

17.3.1 Preprocessing Pipeline for Crop Price Prediction

The information utilized to forecast price trends was gathered from Directorate of Marketing and Inspection's website that comes under the Department of Agriculture and Cooperation, Ministry of Agriculture and Farmers Welfare, Government of India, and was created by the National Informatics Centre [18]. This website provided valuable information to farmers, consumers, traders, and researchers as well, like reliable information regarding the arrival of goods and prices of cereals, pulses, vegetables, and fruits at all the districts' markets of India. The time-series crop price dataset was read from a comma separated values (CSV) file using `pd.read_csv()`. Key columns like the date, meaning the observation time, were changed to a `DateTime` object, and the price, meaning the observed

price of the crop, was used as required for proper forecasting. For data quality, unused features were removed and missing values were handled using forward and backward fill and outlier detection and removal methods. To resolve seasonality effects, new date attributes such as month, quarter, and day of the week were derived. Smoothing techniques such as rolling averages and moving averages were applied to reduce short-run volatility and highlight long-run trends. Feature engineering techniques such as lagged values were also applied to incorporate historical price information, thereby enhancing the model's predictability. The price column was normalized to give a comparable scale for the improved performance of the model. Finally, the data were separated into training and testing datasets according to fixed ratios of 90-10, 80-20, and 70-30.

17.3.2 Implementation of Machine Learning Models for Crop-Price Prediction

Several predictive approaches were employed to leverage their distinctive strengths in crop price forecasting and address the limitations of existing agricultural platforms like Uzhavan app and Kisan Suvidha: ARIMA was adopted as an interpretable baseline to model temporal autocorrelation and generate reliable short- to medium-term forecasts from historical price series; SARIMA extended this framework by incorporating seasonal differencing and seasonal AR/MA terms, thereby capturing periodic price fluctuations driven by planting and harvesting cycles; XGBoost was leveraged to model complex, non-linear interactions among crop prices and exogenous factors (e.g., weather and economic indicators), with built-in regularization and parallel processing ensuring robustness on high-dimensional datasets; random forest was applied to reduce variance through ensemble averaging across decision trees trained on random feature subsets, thus maintaining accuracy amid noisy or partially missing agricultural data; and LSTM networks were introduced to learn both short- and long-term temporal dependencies and abrupt market shifts in volatile time series [19], further enhancing the system's ability to anticipate dynamic price movements. These models were specifically selected to overcome the scalability challenges and computational limitations identified in current agricultural platforms, providing the sophisticated predictive analytics capabilities required for real-time decision support systems that can guide farmers in crop selection, production planning, and market timing strategies.

Table 17.1 shows the characteristics and performance of ARIMA, SARIMA, XGBoost, random forest, and LSTM models.

Table 17.1 Characteristics and performance of machine learning models.

Property	ARIMA	SARIMA	XGBoost	Random forest	LSTM
Type	Statistical	Statistical	Ensemble learning	Ensemble learning	Deep learning
Performance	Good for linear trends and short-term forecasts	Better than ARIMA for seasonal data	High performance for complex, non-linear data	High performance for non-linear data	Excellent for sequential and long-term data
Nature	Deterministic and linear	Deterministic with seasonal components	Non-linear and adaptive	Non-linear and robust	Non-linear and sequential
Handles Seasonality?	No	Yes	No	No	No
Handles non-linear data?	No	No	Yes	Yes	Yes
Strengths	Simple, interpretable and effective for linear trends	Handles seasonality and trends effectively	Captures complex relationships; handles large datasets and additional features	Reduces overfitting; handles missing data and complex interactions	Captures long-term dependency, trends, and seasonality
Best use case	Trends	Seasonality	Features	Interactions	Sequences

17.3.3 Implementation of Ensemble Learning Models for Crop-Price Prediction

Stacked Ensemble Model using LSTM and XGBoost: In our implementation of the hybrid stacked ensemble model with linear regression meta-learner for crop price prediction, we developed a multi-tiered forecasting architecture that combines statistical, ML, and deep learning approaches. At the base level, we configured an ARIMA model with parameters optimized through grid search and Akaike information criterion (AIC) minimization (typically $p = 2$, $d = 1$, and $q = 2$) to capture the linear trends and seasonal patterns inherent in agricultural price time series. Simultaneously, we trained a random forest regressor with 150 estimators, max depth of 15, and min samples split of 5 to model the complex non-linear relationships between features such as rainfall, temperature, production volume, and

previous price points. The third base model, an LSTM network with 64 memory units and tanh activation, was structured with a sequence length of 12 time steps to identify long-term dependencies and recurring patterns in the crop price data. Each model was trained independently on the pre-processed dataset, with appropriate transformations for each algorithm: differencing for ARIMA, feature scaling for random forest, and sequence reshaping for LSTM. The outputs from these three models were then fed as input features to a linear regression meta-learner with L2 regularization (ridge regression, $\alpha = 0.1$) that learned optimal linear coefficients to combine the predictions. This approach leveraged the mathematical simplicity and interpretability of linear regression while minimizing the risk of overfitting at the meta level. The linear meta-model effectively assigned different weights to each base model's predictions based on their historical performance patterns, creating a weighted ensemble that could adapt to different market conditions by relying more heavily on ARIMA during stable trends, random forest during periods with strong feature correlations, and LSTM during complex sequential patterns. This hybrid architecture with linear regression meta-learning provided a computationally efficient yet powerful framework for agricultural price forecasting that balanced model complexity with interpretability.

Hybrid Stacked Ensemble Model: In our implementation of the hybrid stacked ensemble model with XGBoost meta-learner for crop price prediction, we engineered a sophisticated multi-model framework that leverages the complementary strengths of diverse forecasting approaches. The foundation consisted of three base models: an ARIMA model ($p = 2$, $d = 1$, and $q = 2$) with seasonal components ($P = 1$, $D = 1$, $Q = 1$, and $s = 12$) to capture both short-term auto correlations and annual seasonality patterns typical in agricultural markets; a random forest regressor configured with 200 estimators, bootstrap sampling, and feature importance-based selection that excelled at modeling complex interactions between agricultural variables like weather patterns, supply volumes, and market indicators; and an LSTM network with two bidirectional layers of 128 units each and dropout regularization (0.3) designed to capture both forward and backward dependencies in the price sequence data. After independent training and validation of these base models, we collected their predictions on the validation set and constructed a meta-feature matrix where each row represented a time point and columns contained the three model predictions along with additional temporal features indicating market phases. This enriched feature set was then fed into an XGBoost meta-learner (100 estimators, max depth = 4, learning rate = 0.05, and subsample = 0.8) that

was trained to learn non-linear combination rules for integrating the base predictions. The XGBoost meta-model's tree-based architecture enabled it to discover complex conditional relationships effectively learning when to trust each base model under specific market circumstances and how to combine their predictions non-linearly. This approach allowed the ensemble to adapt dynamically to changing market conditions, giving more weight to ARIMA during stable periods with clear seasonality, prioritizing random forest when external factors strongly influenced prices, and favoring LSTM during periods with complex sequential patterns.

Table 17.2 shows the characteristics and performance of stacked ensemble (LSTM + XGBoost), hybrid stacked ensemble (ARIMA + RF + LSTM

Table 17.2 Characteristics and performance of ensemble models.

Property	Stacked ensemble (LSTM + XGBoost)	Hybrid stacked ensemble (ARIMA + RF + LSTM + linear regression)	Hybrid stacked ensemble (ARIMA + RF + LSTM + XGBoost)
Base models	LSTM, XGBoost	ARIMA, random forest, LSTM	ARIMA, random forest, LSTM
Meta-model	Combines sequential learning (LSTM) and non-linear feature modeling (XGBoost)	Combines trend/seasonality (ARIMA), non-linear patterns (RF), and sequential learning (LSTM)	Combines trend/seasonality (ARIMA), non-linear patterns (RF), and sequential learning (LSTM)
Strengths	Combines sequential learning (LSTM) and non-linear feature modeling (XGBoost)	Combines trend/seasonality (ARIMA), non-linear patterns (RF), and sequential learning (LSTM)	Combines trend/seasonality (ARIMA), non-linear patterns (RF), and sequential learning (LSTM)
Handles non-linear data?	Yes (<i>via</i> XGBoost)	Yes (<i>via</i> random forest and LSTM)	Yes (<i>via</i> random forest and LSTM)

+ linear regression), and hybrid stacked ensemble (ARIMA + RF + LSTM + XGBoost) models for crop price prediction.

17.4 Experimental Results

In the experimental results section, we evaluate and compare the performance of individual models (ARIMA, SARIMA, XGBoost, random forest, and LSTM) and ensemble approaches (stacked ensemble with LSTM + XGBoost and hybrid stacked ensembles) for crop price prediction across different data splits, analyzing their ability to capture price trends and patterns through both quantitative metrics and visual comparisons of predicted versus actual prices.

17.4.1 Performance Comparison of ML Models in Crop Price Prediction

ARIMA: Figures 17.2.a, 17.2.b, and 17.2.c have illustrated the comparison between the real and predicted crop prices using the ARIMA model for 70-30, 80-20, and 90-10 dataset splits. The x-axis has represented years (1984–2016), and the y-axis has indicated price. Real prices have been shown as a solid blue line, whereas predicted prices have appeared

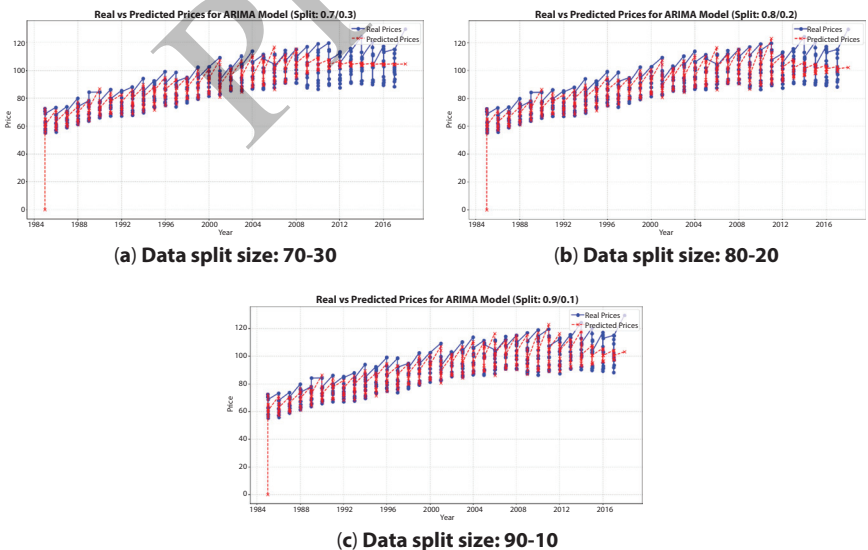


Figure 17.2 Real vs. predicted prices for ARIMA model.

as a dashed orange line. In Figures 17.2.a and 17.2.b, predicted prices have closely followed real values, indicating a good fit. The 90-10 split has resulted in slightly higher deviations due to less training data but has maintained reasonable accuracy. Markers have highlighted key trends, and accuracy has improved as training data has increased, making ARIMA a reliable choice for crop price prediction.

SARIMA: In Figures 17.3.a, 17.3.b, and 17.3.c, the graphs present a comparison between the real and predicted crop prices using the SARIMA model for different dataset splits: 70-30, 80-20, and 90-10. The x-axis spans from 1984 to 2016, whereas the y-axis represents the price. Real prices are illustrated with a solid green line, and predicted prices are shown with a dotted red line, making it easy to distinguish between the two. In Figure 17.3.a, the predicted prices generally track the real prices well, although slight discrepancies are visible in some years. Figure 17.3.b demonstrates a stronger alignment, with the dotted red line closely following the solid green line, indicating improved accuracy. Data points are marked with small circles, highlighting specific trends and variations. Overall, the SARIMA model proves to be effective in predicting crop prices, with performance enhancing as the proportion of training data increases, making it a robust tool for price forecasting in agricultural contexts.

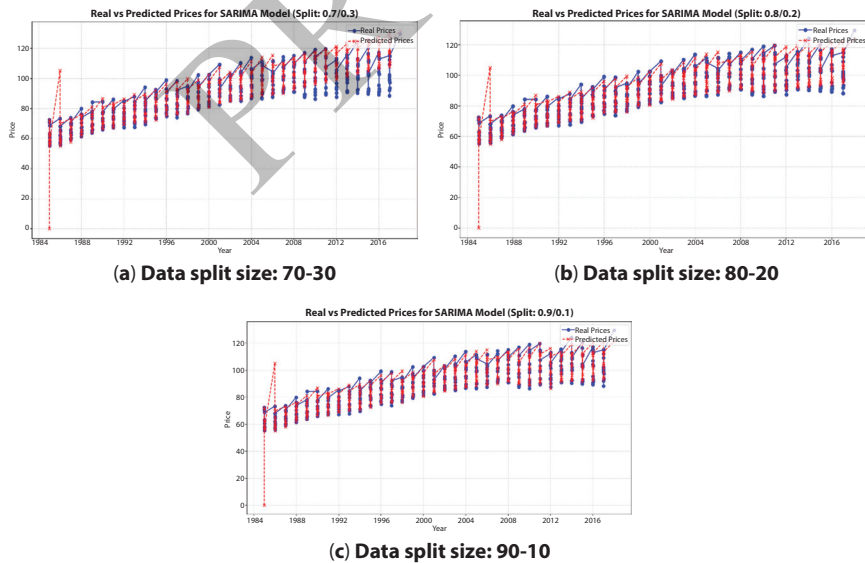


Figure 17.3 Real vs. predicted prices for SARIMA model.

XGBoost: Figures 17.4.a, 17.4.b, and 17.4.c compare the real and predicted crop prices using the XGBoost model for dataset splits of 70-30, 80-20, and 90-10, with the x-axis representing years from 1984 to 2016 and the y-axis indicating price. Real prices are depicted by a solid purple line, whereas predicted prices are shown with a dashed yellow line, ensuring clear differentiation. In Figure 17.4.a, the predicted prices generally align well with the real prices, although some minor deviations are noticeable in certain years. Figure 17.4.b shows improved accuracy, with the dashed yellow line closely following the solid purple line, indicating better model performance. Figure 17.4.c, exhibits greater precision due to the larger training dataset. Data points are marked with small squares, highlighting specific trends and variations.

Random Forest: Figures 17.5.a, 17.5.b, and 17.5.c present the real vs. predicted crop prices using the random forest model across 70-30, 80-20, and 90-10 splits (1984–2016). Real prices are shown with a solid green line; predicted prices are shown with a dashed red line. In Figure 17.5.a, the model achieves 98.60% training and 97.44% testing accuracy, with slight deviations. Figure 17.5.b shows improved performance—99.05% training and 100% testing accuracy, with close alignment. Figure 17.5.c, using more

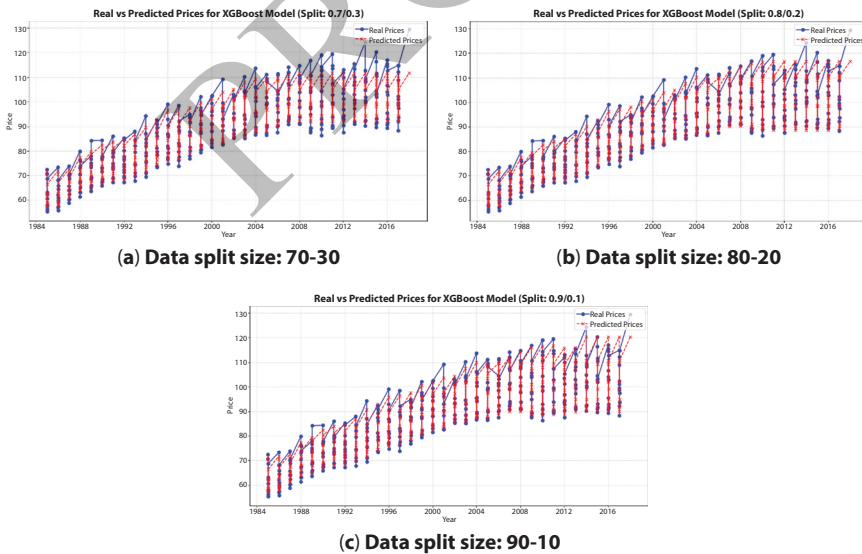


Figure 17.4 Real vs. predicted prices for XGBoost model.

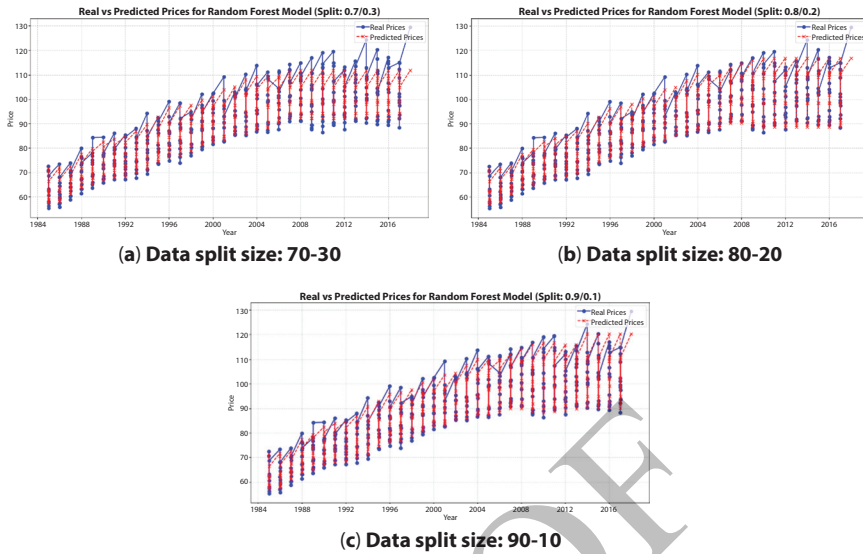


Figure 17.5 Real vs. predicted prices for random forest model.

training data, reaches 98.55% training and 98.32% testing accuracy, confirming the model's reliability and precision.

LSTM: Figures 17.6.a, 17.6.b, and 17.6.c compare the real and predicted crop prices using the LSTM model for dataset splits of 70-30, 80-20, and 90-10, with the x-axis representing years from 1984 to 2016 and the y-axis indicating price. The real prices are depicted by a solid green line, whereas the predicted prices are shown with a dotted red line, making it easy to distinguish between the two. In Figure 17.6.a, the predicted prices generally track the real prices well, although slight discrepancies are visible in some years. Figure 17.6.b demonstrates a stronger alignment, with the dotted red line closely following the solid green line, indicating improved accuracy. Data points are marked with small circles, highlighting specific trends and variations. Overall, the LSTM model proves to be effective in predicting crop prices, with performance enhancing as the proportion of training data increases, making it a robust tool for price forecasting in agricultural contexts.

Table 17.3 shows the performance evaluation of ML models such as ARIMA, SARIMA, random forest, LSTM, and XGBoost with different dataset splits like 70-30, 80-20, and 90-10, comparing their training and testing accuracy for crop price prediction.

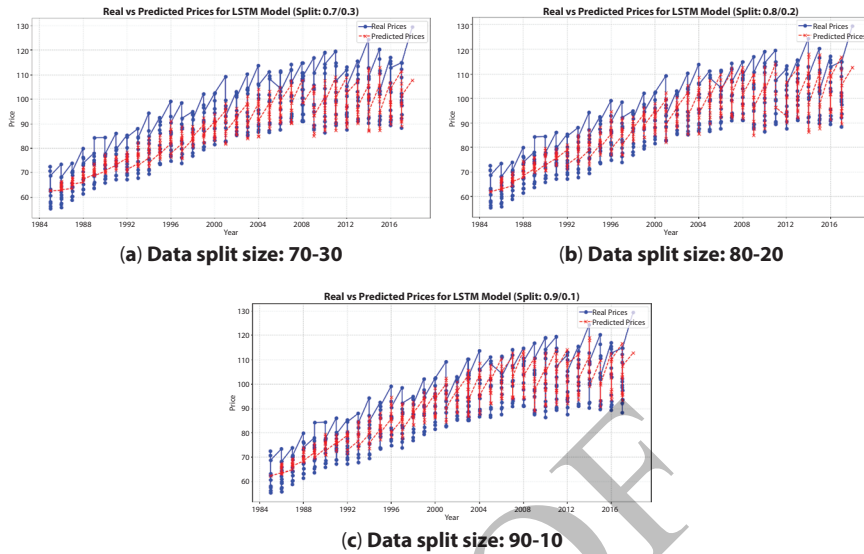


Figure 17.6 Real vs. predicted prices for LSTM model.

Table 17.3 Performance evaluation of machine learning models with different dataset split.

Model name	Dataset split	Training accuracy	Testing accuracy
ARIMA	0.7, 0.3	96.22	92.78
	0.8, 0.2	96.29	94.85
	0.9, 0.1	96.37	94.37
SARIMA	0.7, 0.3	91.67	91.60
	0.8, 0.2	91.45	89.87
	0.9, 0.1	90.45	94.87
Random forest	0.7, 0.3	98.60	97.44
	0.8, 0.2	98.05	97.63
	0.9, 0.1	98.55	98.32
LSTM	0.7, 0.3	97.62	94.63
	0.8, 0.2	97.71	96.67
	0.9, 0.1	98.60	96.85
XGBoost	0.7, 0.3	97.71	86.84
	0.8, 0.2	96.78	92.20
	0.9, 0.1	97.43	87.07

17.4.2 Performance Comparison of Ensemble Learning Models in Crop Price Prediction

Stacked Ensemble Model using LSTM and XGBoost: Figures 17.7.a, 17.7.b, and 17.7.c compare the real vs. predicted crop prices using a stacked ensemble model (LSTM + XGBoost) across dataset splits of 70-30, 80-20, and 90-10 (1984–2016). Real prices are shown as a solid blue line, and predicted prices are shown as a dashed red line. The 70-30 split shows strong predictive alignment; the 80-20 split yields even closer correlation, whereas the 90-10 split achieves the most accurate forecasts due to more training data. Highlighted data points capture key trends, validating the model's robustness in crop price prediction.

Hybrid Stacked Ensemble Model - Linear: Figures 17.8.a, 17.8.b, and 17.8.c compare the real and predicted crop prices using the stacked ensemble model with linear regression as the meta-model for dataset splits of 70-30, 80-20, and 90-10. The x-axis represents years (1984–2016), whereas the y-axis indicates price. Real prices are shown by a solid blue line, and predicted prices are shown by a dashed red line for clear differentiation. For the 70-30 split, predicted prices closely follow real prices with minor deviations, indicating a strong fit. The 80-20 split exhibits improved alignment, showcasing enhanced predictive accuracy. The 90-10 split, despite

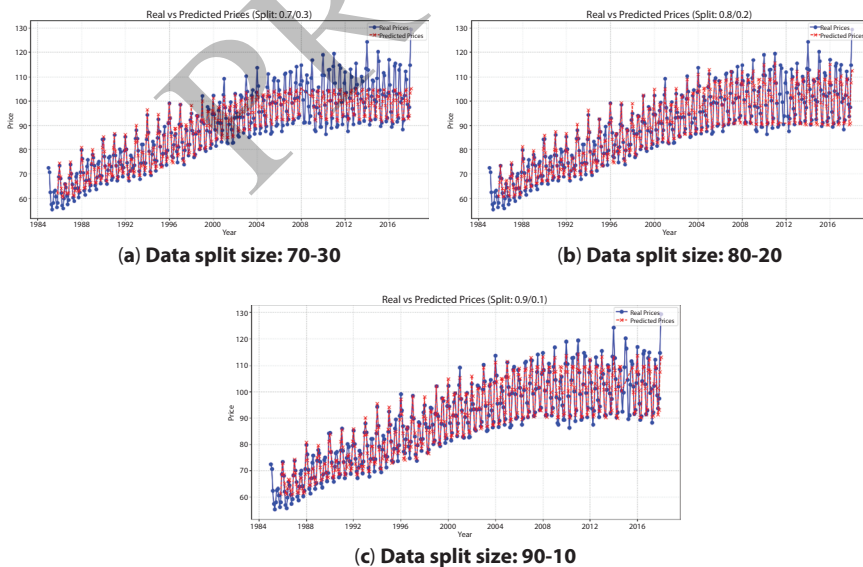


Figure 17.7 Real vs. predicted prices for stacked ensemble model using LSTM and XGBoost.

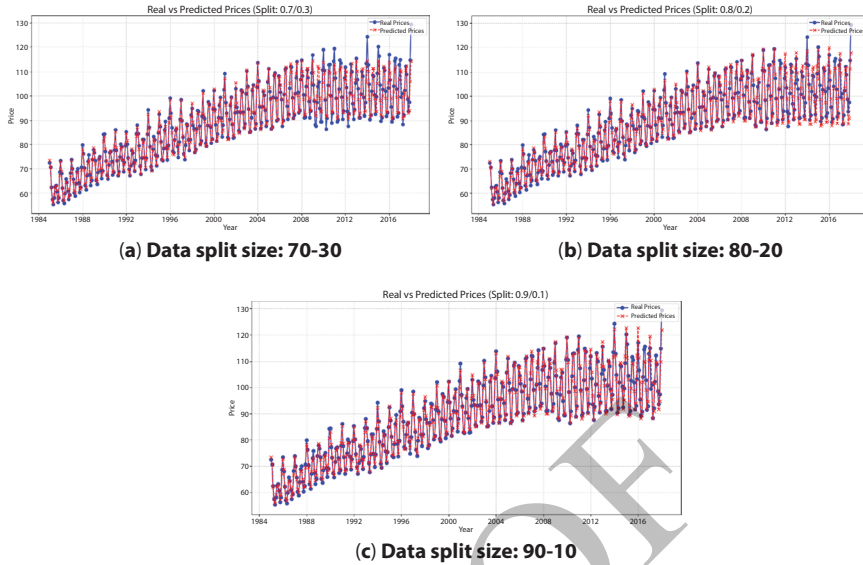


Figure 17.8 Real vs. predicted prices for hybrid stacked ensemble model - linear.

using less training data, maintains reasonable precision, although deviations are slightly higher. These results demonstrate the stacked ensemble model's effectiveness in leveraging ARIMA, random forest, and LSTM to enhance forecasting accuracy, with linear regression optimizing the final predictions.

Hybrid Stacked Ensemble Model: Figures 17.9.a, 17.9.b, and 17.9.c depict the comparison between the real and predicted crop prices using the ARIMA model for dataset splits of 70-30, 80-20, and 90-10, with the x-axis representing years from 1984 to 2016 and the y-axis indicating price values. The real prices are shown with a solid line, whereas the predicted prices are represented by a dashed line. For the 70-30 split, the predicted prices closely follow the real prices, with minor deviations in certain years, indicating a good fit. The 80-20 split shows a similar trend, with the predicted prices aligning well with the real prices, suggesting consistent model performance. The 90-10 split exhibits closer alignment due to the larger training dataset. Overall, the ARIMA model demonstrates strong predictive capability for crop prices, with accuracy improving as the training dataset size increases. This suggests that ARIMA is a reliable model for crop price prediction, especially when a substantial portion of the data is used for training.

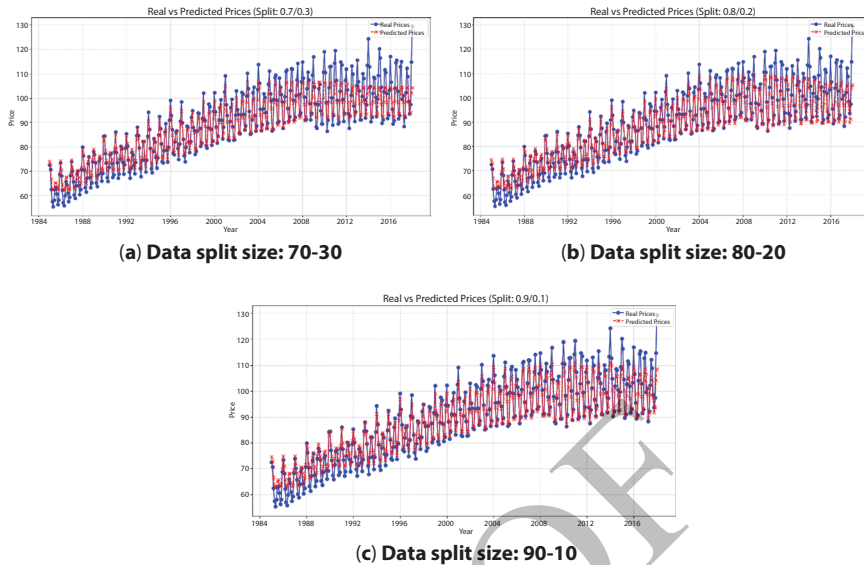


Figure 17.9 Real vs. predicted prices for hybrid stacked ensemble model - XGBoost.

Table 17.4 shows performance evaluation of stacked ensemble (LSTM + XGBoost), hybrid stacked ensemble (ARIMA + RF + LSTM + linear regression), and hybrid stacked ensemble (ARIMA + RF + LSTM + XGBoost) models with different dataset splits like 70-30, 80-20, and 90-10, comparing their training and testing accuracy for crop price prediction.

Table 17.4 Performance evaluation of ensembled models with different dataset splits.

Models	Dataset split size	Training data accuracy	Testing data accuracy
Stacked ensemble (LSTM + XGBoost)	0.7, 0.3	97.68	95.35
	0.8, 0.2	98.12	96.94
	0.9, 0.1	98.03	97.24
Hybrid stacked ensemble (ARIMA + RF + LSTM + linear regression)	0.7, 0.3	98.13	97.22
	0.8, 0.2	98.08	96.89
	0.9, 0.1	99.01	97.24
Hybrid stacked ensemble (ARIMA + RF + LSTM + XGBoost)	0.7, 0.3	97.65	95.59
	0.8, 0.2	97.60	95.88
	0.9, 0.1	97.58	95.93

17.5 Conclusions and Future Work

This research successfully evaluated various ML models and ensemble techniques for crop price prediction. Our comprehensive analysis revealed that tree-based models, particularly random forest, consistently outperformed traditional statistical approaches, achieving up to good testing accuracy with an 80-20 split. LSTM models demonstrated strong performance (94.43%–96.85% testing accuracy) by effectively capturing temporal dependencies, whereas XGBoost showed high training accuracy but more variable testing performance, suggesting potential overfitting. Traditional statistical models like ARIMA (92.78%–94.85%) and SARIMA (89.87%–94.87%) performed adequately but were consistently outperformed by ML approaches, confirming their limitations with complex, non-linear patterns in agricultural price data. The hybrid stacked ensemble model further enhanced prediction accuracy by leveraging the complementary capabilities of multiple algorithms.

Future research should focus on: (1) integrating external factors such as weather conditions, soil quality, and global market dynamics; (2) developing real-time prediction systems with continuous updates; (3) implementing explainable AI techniques to increase transparency; (4) exploring transfer learning for low-resource agricultural regions; (5) creating commodity-specific models that account for unique growth cycles and market dynamics; (6) investigating blockchain integration for secure data sharing; (7) extending models to provide multi-horizon forecasting; (8) exploring more sophisticated hybrid deep learning architectures; (9) developing models to simulate policy interventions; and (10) creating user-friendly mobile applications for farmers with limited technical expertise.

The methodology developed in this research provides a comprehensive framework for crop price prediction that can be adapted to different agricultural commodities and market conditions. These findings contribute significantly to agricultural forecasting methods and offer valuable tools for farmers, stakeholders, and policymakers to make informed decisions in crop marketing and management, ultimately contributing to improved food security and economic stability in the agricultural sector.

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References

1. Kaewchada, S. and Ruang-On, S., Random forest model for forecasting vegetable prices. *Int. J. Electr. Comput. Eng. (IJECE)*, 13, 6, 5265–5272, 2023.
2. Samuel, P., Crop Price Prediction System using Machine Learning Algorithms. *Quest J. - J. Softw. Eng. Simul.*, 6, 1, 1–13, 2020.
3. Singh, N., Crop Price Prediction Using Machine Learning. *J. Electr. Syst.*, 10, 2, 2258–2269, 2024.
4. Soni, N., Crop Price Prediction Using Machine Learning Techniques. *Int. Adv. Res. J. Sci. Eng. Technol.*, 9, 3, 1–10, 2022.
5. Murugesan, G., An extrapolative model for price prediction of crops using hybrid ensemble learning techniques. *Int. J. Adv. Technol. Eng. Explor.*, 10, 4, 1–10, 2023.
6. Liakos, K., Busato, P., Moshou, D., Pearson, S., Bochtis, D., Machine learning in agriculture: a review. *Sensors*, 18, 8, 1–20, Aug. 2018.
7. Rohit, R., Vishnu, R., Kishore, A., Chakkarawarthy, D., Crop Price Prediction and Forecasting System using Supervised Machine Learning Algorithms. *Int. J. Adv. Res. Comput. Commun. Eng. (IJARCCE)*, 9, 3, 1–8, 2020.
8. Ghutake, I., Verma, R., Chaudhari, R., Amarsinh, V., An intelligent Crop Price Prediction using suitable Machine Learning Algorithm, in: *Proceedings of the International Conference on Advances in Computing and Communication (ICACC-2021)*, ITM Web of Conferences, vol. 40, pp. 1–7, 2021.
9. Zhang, D., Chen, S., Liwen, L., Xia, Q., Forecasting agricultural commodity prices using model selection framework with time series features and forecast horizons. *IEEE Access*, 8, 28197–28209, 2020.
10. Subhasree, M. and Priya, C.A., Forecasting vegetable price using time series data. *Int. J. Adv. Res. (IJAR)*, 3, 535–641, 2016.
11. Bayona-Oré, S., Cerna, R., Hinojoza, E.T., Machine learning for price prediction for agricultural products. *WSEAS Trans. Bus. Econ.*, 18, 969–977, 2021.
12. Sahithi, B., Saheli, T., Ramanika, D., Anil Kumar, N., Crop Price Prediction and Forecasting System using Supervised Machine Learning Algorithms. *J. Softw. Eng. Simul.*, 6, 1, 14–20, 2020.
13. Kaur, M., Gulati, H., Kundra, H., Data Mining in Agriculture on Crop Price Prediction: Techniques and Applications. *Int. J. Comput. Appl.* (0975 – 8887), 99, 12, 1–3, 2014.
14. Mukhtar, M., *et al.*, Hybrid model in machine learning–robust regression applied for sustainability agriculture and food security. *Int. J. Electr. Comput. Eng. (IJECE)*, 12, 4, 4457–4468, Aug. 2022.
15. Sabu, K.M. and Kumar, T.K.M., Predictive analytics in Agriculture: Forecasting prices of Arecanuts in Kerala. *Procedia Comput. Sci.*, 171, 2231–2240, 2020.
16. Darekar, A. and Reddy, A.A., Forecasting of Common Paddy Prices in India. *J. Rice Res.*, 10, 1, 35–42, 2017.

17. Sun, F., Meng, X., Zhang, Y., Wang, Y., Jiang, H., Liu, P., Agricultural Product Price Forecasting Methods: A Review. *Agriculture*, 13, 9, 1–23, 2023.
18. Department of Marketing & Inspection, Government of India. [Online]. Available: <https://dmi.gov.in/>.
19. Géron, A., *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 2nd edn., O'Reilly Media, Sebastopol, CA, USA, 2019.
20. Jayasri, P.A., Manimegalai, R., Reshmah, C.S., Vaishnavi, S., Evaluation of Weather Forecasting Models and Handling Anomalies in Short-Term Wind Speed Data, in: *Advances in Distributed Computing and Machine Learning*, Lecture Notes in Networks and Systems, vol. 955, Springer, 2024.
21. Kingsy Grace, R., Madhumathi, M., Manimegalai, R., Efficient Data Prediction Scheme Using Time Series Classification. *Second International Conference on Innovative Research in Engineering and Technology*, India, January, 2013.
22. Kingsy Grace, R. and Manimegalai, R., Design of Neural Network based Wind Speed Prediction model using GWO. *Comput. Syst. Sci. Eng.*, 40, 2, 593–606, 2022.